

Research article

NH₃, N₂O, NO and NO₂ losses, emission factors and reduction mechanisms under different N managements from greenhouse vegetable soils

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ABSTRACT

Excessive nitrogen (N) application in greenhouse cultivation has led to substantial reactive N (Nr) emissions from soils, necessitating effective mitigation strategies. This study presents simultaneous high-frequency monitoring of ammonia (NH₃) volatilization, nitrous oxide (N₂O), nitric oxide (NO) and nitrogen dioxide (NO₂) emissions across three greenhouse vegetable seasons (tomato, cabbage and lettuce) under different N management regimes. Key findings reveal that emission factors of NH₃, N₂O, NO, and NO₂ were 0.6 %, 1.6 %, 0.8 % and 0.2 % in greenhouse vegetable systems, respectively. Among the emitted annual Nr gases (8.2–72.0 kg N-ha⁻¹), N₂O emissions accounted for the highest proportion of total Nr gas losses. Controlled-release N with organic fertilizer emerged as the most effective strategy, significantly reducing total Nr gas emissions while maintaining high vegetable yield. This reduction was mechanistically linked to suppressed urease activity and decreased expression of key microbial genes (*amoA*, *hao*, *nirK* and *norB*). N application rate, fertilizer type, soil moisture and temperature were identified as the primary drivers of Nr gas emissions. This study firstly provided crucial empirical emission factors and proportions for NH₃, N₂O, NO and NO₂ gases, offering a technical pathway for synergistically reducing pollution and carbon emissions in greenhouse vegetable cultivation.

1. Introduction

Greenhouse cultivation, since 1980, has effectively addressed the challenge of ensuring a sustainable supply of vegetables, resulting in a significant expansion (Li et al., 2022). By 2022, China's greenhouse horticulture area accounted for over 80 % of the global total, covering an area of 2.8 million hectares (Li et al., 2022). Moreover, the economic benefits derived from greenhouse vegetable cultivation are estimated to be 4–5 times higher than those for open-field cultivation (Li et al., 2022). Aiming to pursue high yield and income, excessive chemical fertilizers are applied in greenhouses by farmers. Nitrogen (N) application rates for vegetables surpass the required quantity almost 3–5 times (Huang et al., 2016; Ju et al., 2007; Zhang et al., 2018). The inappropriate application

of N fertilizer caused negative cascading effects on the environment, particularly regarding reactive nitrogen (Nr) gas losses (Yi et al., 2023).

The losses of Nr gas mainly encompass ammonia (NH₃) volatilization, nitrous oxide (N₂O), nitric oxide (NO) and nitrogen dioxide (NO₂) emissions (Ding et al., 2023). N₂O emission contributes to the greenhouse effect and climate warming. NH₃ serves as a precursor substance for PM_{2.5}, leading to an increase in PM_{2.5} levels and causes indirect N₂O emissions. NO is a key tropospheric oxidant, contributing to tropospheric ozone formation and regional haze, with cascading impacts on human health (e.g., the formation of regional haze and outbreak of respiratory and cardiovascular diseases) and ecosystems. Critically, atmospheric NO and NO₂ deposition mediates soil acidification through HNO₃ generation, and aquatic eutrophication via nitrate leaching that

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triggers algal blooms and hypoxia (Qin et al., 2017; Wang et al., 2022). Investigating approaches to reduce Nr gas emissions is therefore crucial and urgent to alleviate both pollution and high carbon emissions in horticulture and achieve sustainable agricultural production.

The determination of Nr gas emission characteristics, emission factors (EFs) and emission proportions from greenhouse vegetable soil is challenging due to the lack of simultaneous observation of NH_3 , N_2O , NO and NO_2 . This limits an accurate assessment of the multidimensional impacts of NH_3 , N_2O , NO and NO_2 on environmental pollution, human health and global warming. Numerous studies have focused on N management strategies to reduce NH_3 and N_2O emissions, while limited attention has been paid to NO and NO_2 emissions, thus the EFs of NO and NO_2 are with high level of uncertainty. Organic N fertilizer is usually applied to greenhouse vegetable cultivation combined with inorganic N fertilizer (Yan et al., 2023; Xu et al., 2024). The effect of combined application of organic and inorganic fertilizer on N_2O emissions is inconsistent (Shuai et al., 2018; Wang et al., 2016). But the combination of organic and inorganic fertilizer results in lower NH_3 volatilization (Miao et al., 2022), and there is a gradually decreased trend in NH_3 volatilization with the increase ratio of organic N rate (Hao et al., 2012). Application of controlled-release N fertilizer has demonstrated a significant reduction in N_2O emissions by 38.1–47.0 % and NH_3 volatilization by 78–83 % compared to conventional fertilizer (Shan et al., 2015; Yang et al., 2014). While little is known about the effects of organic fertilizer substitution and controlled-release N fertilizer on NO and NO_2 emissions in greenhouses. It is worthwhile investigating the potential for reducing Nr gas emission by organic fertilizer substitution and controlled-release fertilizer application in greenhouse vegetable cultivation.

Furthermore, environmental factors, soil properties and microbes, such as temperature, moisture, soil available N content, enzyme activity, nitrifier and denitrifier, have vital impacts on N cycling in greenhouses, thus affecting Nr emissions (Bi et al., 2024; Kong et al., 2021). Specifically, hydroxylamine reductase and denitrifying enzymes are pivotal catalysts in soil nitrogen metabolism, mediating nitrate formation and the emission of N_2O , NO, and NO_2 (Deng et al., 2024; Wang et al., 2024). The production and consumption of these gases are tightly regulated by functional genes encoding key enzymes involved in nitrification (*amoA* for ammonia oxidation, *hao* for hydroxylamine oxidation) and denitrification (*narG* for nitrate reduction, *nirS/nirK* for nitrite reduction, *norB* for nitric oxide reduction, and *nosZ* for nitrous oxide reduction) (Deng et al., 2024; Quick et al., 2019). In contrast, NH_3 volatilization primarily arises from the physicochemical hydrolysis of urea, a process catalyzed by urease through substrate-specific enzymatic activity (Wang et al., 2024). With the application of organic-inorganic fertilizer and controlled-release fertilizer in greenhouse, the involved mechanism on NH_3 , N_2O , NO and NO_2 reduction associated with environment, soil and microbe is poorly understood.

Therefore, we implemented modified N fertilizer management strategies in greenhouse vegetable cultivation with a goal of reducing Nr gas emissions compared to conventional fertilizer applications, i.e., increasing the proportion of organic fertilizer in organic-inorganic fertilizer application, all-organic fertilizer application and controlled-release fertilizer replacing an equal amount of urea. NH_3 , N_2O , NO, and NO_2 emissions were synchronously monitored at a high frequency throughout the greenhouse vegetable cultivation. The objectives of this study were to identify: (1) NH_3 , N_2O , NO, and NO_2 emission EFs and reductions of Nr gas; (2) the involved mechanism of NH_3 , N_2O , NO, and NO_2 reduction; and (3) the contributions of N fertilizer management, environmental factors, and soil properties to NH_3 , N_2O , NO, and NO_2 emissions. This study could facilitate achieving sustainable production in greenhouse vegetable cultivation, providing valuable insights into synergistically reducing pollution and carbon emissions.

2. Materials and methods

2.1. Experiment sites

The experimental station was located in Yanqing District, Beijing, China (40.54°N, 116.07°E) with a warm temperate semi-humid semi-arid monsoon climate. A typical solar greenhouse with 120 m long and 9 m wide was employed, which has continuously cultivated vegetables for 11 years. The physicochemical properties of the soil at depth of 0–20 cm in the greenhouse were: pH (H_2O), 7.7; organic matter (SOM) content, 42.4 g kg^{-1} ; alkaline nitrogen, 178.0 mg kg^{-1} ; rapidly available potassium, 507.0 mg kg^{-1} ; rapidly available phosphorus, 303.1 mg kg^{-1} .

2.2. Experiment design

There were five treatments in this experiment: (1) control (CK, no fertilizer), (2) 25 % organic fertilizer and 75 % chemical fertilizer (25 % OF, conventional fertilization), (3) 50 % organic fertilizer and 50 % chemical fertilizer (50 %OF), (4) 100 % organic fertilizer (100 %OF), (5) 50 % organic fertilizer and 50 % controlled-release fertilizer (50 %CRF). The experiment was conducted with three replications in a randomized block design. The field experiment began from 2020 to 2021, sequentially cultivating tomato, cabbage and lettuce in the same plot. The application rates of phosphate and potassium fertilizer were: 738 kg $\text{P}_2\text{O}_5\cdot\text{ha}^{-1}$ and 486 kg $\text{K}_2\text{O}\cdot\text{ha}^{-1}$ in tomato cultivation, 277 kg $\text{P}_2\text{O}_5\cdot\text{ha}^{-1}$ and 182 kg $\text{K}_2\text{O}\cdot\text{ha}^{-1}$ in cabbage cultivation, 138 kg $\text{P}_2\text{O}_5\cdot\text{ha}^{-1}$ and 91 kg $\text{K}_2\text{O}\cdot\text{ha}^{-1}$ in lettuce cultivation. The N application rates were equal except CK treatment, which were shown in Table 1.

Each plot was 7.2 (0.8 m $\times$ 9 m) m^2 , and tomato was cultivated in 2 rows per row, while cabbage and lettuce were cultivated in 3 rows per row. The basal fertilizer of all the five treatments was solid organic fertilizer (OF, chicken manure, $\text{N} \geq 1.26\%$); controlled-release fertilizer (CRF, $\text{N} \geq 42\%$) for 50 %CRF was also applied as basal fertilizer; the topdressing for 25 %OF and 50 %OF was urea (CF, 46 % N), the topdressing for 100 %OF was liquid vermicompost (80.84 g $\text{N}\cdot\text{L}^{-1}$). The frequency of topdressing depends on the growth conditions. It was applied 5 times in tomato cultivation, 3 times in cabbage cultivation, and 1 time in lettuce cultivation. Field preparation, tillage, drip irrigation, and the application of pesticide or herbicide were carried out manually in accordance with local farming practices.

2.3. Measurements of Nr gas emissions

2.3.1. Measurement of NH_3 volatilization

NH_3 volatilization was measured by venting method (Wang et al., 2002). The device was composed of two sponges and a rigid polyvinyl chloride (PVC) plastic pipe. The outer diameter of the base of the PVC pipe is 17 cm (inner diameter is 16 cm), the upper outer diameter is 11 cm (inner diameter is 10 cm), and the height of the device is 12 cm. Each sponge was soaked in 15 mL glycerin phosphate solution (50 mL phosphoric acid and 40 mL glycerol fixed to 1 L), the upper sponge corresponds to the top of the PVC pipe, while the lower sponge is 5 cm from the ground. There were 2 PVC buckets in each replicate, 2 lower sponges as one sample. After absorbing ammonia, each sample was soaked in 300 mL KCL (1.0 mol L^{-1}), and shaken for 1 h (250 r $\cdot\text{min}^{-1}$), before being filtered. The concentration of $\text{NH}_4^+\text{-N}$ in the extract was determined by flow analyzer. Sponges should be replaced on the 1st, 2nd, 3rd, 5th, 7th, 10th and 15th days after fertilization, at 8:00–10:00 a.m. each time, and replace sponges every 5 days until the next fertilization. The NH_4^+ concentration was used for NH_3 flux (F_{NH_3} , $\text{kg}\cdot\text{ha}^{-1}\cdot\text{d}^{-1}$) calculation by Eq. (1):

$$F_{\text{NH}_3} = \frac{M}{A \times D} \times 10^{-2} \quad (1)$$

where M is the NH_4^+ concentration of each sample (mg), A is the lower

Table 1

Application rate of N fertilizers in greenhouse tomato, cabbage and lettuce cultivation.

Treatments	Tomato (kg N·ha ⁻¹)		Cabbage (kg N·ha ⁻¹)		Lettuce (kg N·ha ⁻¹)	
	Organic	Inorganic	Organic	Inorganic	Organic	Inorganic
CK	0	0	0	0	0	0
25 %OF	200	600	75	225	37.5	112.5
50 %OF	400	400	150	150	75	75
100 %OF	800	0	300	0	150	0
50 %CRF	400	400	150	150	75	75

cross-sectional area of the device (m²), and D is the duration of each sampling interval (day). After calculating daily NH₃ volatilization fluxes during the growing season for each N management, the 90th percentile value was established as the threshold. Continuous periods exceeding this flux threshold were defined as “emission peak” events.

2.3.2. Measurement of N₂O emission

N₂O emission was measured by a closed static chamber method. The static chamber has a diameter of 25 cm, a height of 50 cm. The base was inserted into the soil at 9 cm, and 12 cm above ground as a pedestal for the chamber. Each sampling took 30 min. The gas sample was manually extracted from the chamber through a needle into a vacuum tube at 0, 10, 20, and 30 min after mounting the static chamber on the base with a water seal, and the gas should be mixed before sampling. The gas samples were collected in order to determine the concentration of N₂O using a gas chromatograph (Agilent 7890A, USA) within 24 h. The chromatographic conditions referred to Wang et al. (2018). The gas flux was calculated by changes of the 4 samples. The result was valid when the linear regression value of R² was greater than 0.8. Samplings were taken and measured on the 1st, 2nd, 3rd, 5th, 7th, 10th and 15th days after fertilization, at 9:00–11:00 a.m. each time, and measured every 5 days until the next fertilization. The change of the samples was used for N₂O flux (F_{N2O}, mg·m⁻²·h⁻¹) calculation by Eq. (2):

$$F_{N_2O} = \rho \times \frac{V}{A} \times \frac{dc}{dt} \times \frac{273}{T} \quad (2)$$

where ρ is the density of N₂O in the standard state (kg·m⁻³), V is the effective volume of the static chamber (m³), A is the soil area covered by the base (m²), dc/dt is the change of N₂O concentration of 4 samples in unit time (uL·L⁻¹·h⁻¹), T indicates the Fahrenheit temperature in the chamber (K). After calculating daily N₂O emission fluxes during the growing season for each N management, the 90th percentile value was established as the threshold. Continuous periods exceeding this flux threshold were defined as “emission peak” events.

2.3.3. Measurement of NO and NO₂ emissions

NO and NO₂ levels were measured using the NO-NO₂-NO_x analyzer (Thermo 17i). The NO-NO₂-NO_x analyzer was connected to the static chamber, and the gas from the chamber was fed into the NO-NO₂-NO_x analyzer through a gas line to monitor real-time gas concentration, with one set of data (NO and NO₂) being monitored every 1 min. The static chamber is 0.35 m high, 0.7 m long and 0.35 m wide. The electronic module automatically controls the static chamber switch. When the static chamber is closed, the gas from this chamber flows into the NO-NO₂-NO_x analyzer through the pipeline, the sampling chamber is closed for 15 min for monitoring. This sampling chamber is opened after the measurement is completed, while the gas circuit of this static box is closed, and the next sampling chamber is closed to start the concentration monitoring of the next chamber. Fifteen independent gas circuit pipelines are set up to collect gas monitoring of 5 treatments (3 replicates) in turn. After the last sampling chamber is completed, it will automatically switch to the first sampling chamber, allowing continuous monitoring during vegetable cultivation. The method of NO and NO₂ measurement was based on Ding et al. (2022), except that in their experiments they sampled 100 mL of gas at 0, 6, 12, 18, and 24 min in the

static box and measured it within 1 h, but this experiment got 15 min continuous real-time measurements and selected data from the 11th to the 15th minute for gas flux calculation. The result was valid when the linear regression value of R² was greater than 0.8. The NO and NO₂ emission fluxes (F_{NO/NO2}, mg·m⁻²·h⁻¹) were also calculated by Eq. (3):

$$F_{NO,NO_2} = \rho \times \frac{V}{A} \times \frac{dc}{dt} \times \frac{273}{T} \quad (3)$$

where ρ is the density of NO or NO₂ in the standard state (kg·m⁻³), V is the effective volume of the static chamber (m³), A is the soil area covered by the static chamber (m²), dc/dt is the change of NO or NO₂ concentration in unit time (uL·L⁻¹·h⁻¹), T indicates the Fahrenheit temperature in the static chamber (K). After calculating daily NO and NO₂ emission fluxes during the growing season for each N management, the 90th percentile value was established as the threshold. Continuous periods exceeding this flux threshold were defined as “emission peak” events.

2.4. EF calculation of Nr gases

The EF of Nr gas (NH₃, N₂O, NO and NO₂) was calculated by Eq. (4):

$$EF_n(\%) = \frac{E_{n,Treatment} - E_{n,Control}}{N \text{ application rate}} \times 100 \quad (4)$$

where n is one of the four gases (NH₃, N₂O, NO and NO₂), $E_{n,Treatment}$ is the cumulative gaseous Nr emission of N management (kg N·ha⁻¹), $E_{n,Control}$ is the cumulative gaseous Nr emission of CK (kg N·ha⁻¹).

The Nr gas loss was summed up by the cumulative emission of NH₃, N₂O, NO and NO₂. The EF of Nr gas was calculated by Eq. (5):

$$Nr \text{ gas EF}(\%) = \frac{Nr \text{ gas } N_{(Treatment)} - Nr \text{ gas } N_{(Control)}}{N \text{ application rate}} \times 100 \quad (5)$$

where $Nr \text{ gas } N_{(Treatment)}$ is the Nr gas emission of N management (kg N·ha⁻¹), the $Nr \text{ gas } N_{(Control)}$ is the Nr gas emission of CK (kg N·ha⁻¹).

In Eq. (5), all gaseous emissions (NH₃, N₂O, NO and NO₂) were standardized to kg N·ha⁻¹ units when calculating total Nr gas emissions. Similarly, proportions of individual Nr gases relative to total Nr gas emissions were calculated using consistent N-based units (kg N·ha⁻¹).

2.5. Yield and environmental data

Fruit yields (fresh weight) in each replicate were recorded, and vegetable yields were measured from picking and pulling. The frequency of measuring soil pH (0–20 cm) was the same as that of N₂O measurements (Fig. S1). Data of air temperature, air humidity, soil temperature and soil moisture were monitored automatically, the monitoring probes were calibrated prior to use, and the monitoring frequency was once every 10 min (Figs. S2–S3).

2.6. Soil NO₃⁻-N contents, NH₄⁺-N contents, enzyme activities and quantitative PCR (qPCR)

A sample of soil, depth of 0–20 cm, was taken both before and after each fertilization to measure the levels of nitrate nitrogen (NO₃⁻-N) and

ammonium nitrogen ($\text{NH}_4\text{-N}$), the NO_3^- and NH_4^+ were determined by flow injected analysis method (Fig. S4). Soil enzyme activities and gene copy numbers were measured in the topsoil after the harvest of cabbage (end of the field experiment). The methods used for determining soil enzyme activities were as follows: urease activity was measured using the indophenol blue method, denitrification enzyme activity was analyzed using gas chromatography, and protease, dehydrogenase, and hydroxylamine reductase were measured using a UV spectrophotometer (Hash DR6000, China). All the soil samples were stored at -80°C .

DNA was extracted from soil samples using the FastDNA Spin Kit (MP Biomedicals, USA). The concentration and purity of the extracted DNA were measured with a NanoDrop One/One^c Micro UV-Vis spectrophotometer (Thermo Fisher Scientific, USA). The DNA samples were stored at -80°C for further analysis. Quantitative polymerase chain reaction (qPCR) was conducted in a 20 μL reaction mixture, which included 10 μL of 2X Taq Plus Master Mix, 0.8 μL of forward primer (5 μM), 0.8 μL of reverse primer (5 μM), 1 μL of template DNA, and 7.4 μL of ddH₂O. The qPCR cycling conditions are detailed as: pre-denaturation, 95°C , 5 min; denaturation, 95°C , 30 s; annealing, 72°C , 30 s; elongation, 72°C , 1 min. The cycle number is 35. The qPCR was conducted using gene-specific primers (Table S1) to detect the following genes related to soil N cycling: ammonia monooxygenase (*amoA*), hydroxylamine oxidoreductase (*hao*), nitrate reductase (*narG*), nitrite reductase (*nirS/nirK*), nitric oxide reductase (*norB*), and nitrous oxide reductase (*nosZ*).

2.7. Statistical analysis

Data processing and calculations were done with Microsoft Excel 2016. Graphpad Prism 7.0 and SPSS 22.0 were used for data analysis, including analysis of variance (ANOVA), *t*-test and multiple comparisons. The least significant difference (LSD) test was employed for multiple comparisons. The prediction of gas emission contribution was performed using a gradient boosted regression tree model (BRT analysis). Driving factors included in this experiment were: N application

rate, N fertilizer type, air temperature, air humidity, soil temperature, soil moisture, soil NO_3^- -N content, soil NH_4^+ -N content and soil pH. The parameters used in BRT analysis were: learning rate (0.01), bag fraction (0.50), cross-validation and tree-complexity (Cai et al., 2022; Elith et al., 2008). The BRT model was implemented in the GBM package in R version 3.3.3 (Elith et al., 2008). All figures in this paper were drawn with Originlab 2021.

3. Results

3.1. Emission fluxes and cumulative emissions of NH_3 , N_2O , NO and NO_2

Significant variations in NH_3 , N_2O , NO and NO_2 emission patterns were observed across different fertilization treatments. NH_3 , NO and NO_2 fluxes of 25 %OF were significantly higher than other treatments during the topdressing period, while 100 %OF achieved the highest N_2O flux (Fig. 1). The NH_3 and N_2O emission peaks lasted 3–8 days after basal fertilization and topdressing (Fig. 1A and B), while the NO and NO_2 emission peak continued for 3–15 days (Fig. 1C and D). Compared with 25 %OF (conventional fertilizer), the cumulative NH_3 volatilizations were 12.7, 3.6, and 1.7 kg N-ha⁻¹ for tomato, cabbage and lettuce cultivation, respectively. Compared with this baseline, reductions of 40.5 %, 23.3 % and 12.8 % were observed under 50 %OF (organic substitution), and 61.9 %, 49.4 % and 23.3 % under 50 %CRF (controlled-release fertilizer) (Fig. 2A–C, Table S2). The highest N_2O emission was observed under 100 %OF, with values around 23.2, 6.2 and 2.5 kg N-ha⁻¹ for tomato, cabbage and lettuce cultivation, respectively, compared to 25 %OF baseline values of 18.3, 5.6, and 2.0 kg N-ha⁻¹. For NO emissions, the NO emissions of 25 %OF were 16.2, 5.5, and 1.9 kg N-ha⁻¹ for tomato, cabbage and lettuce. Compared with this baseline, reductions of 51.7 %, 61.4 % and 64.2 % occurred under 50 %OF, and 87.2 %, 79.5 % and 66.7 % under 50 %CRF (Fig. 2A–C, Table S2). Additionally, the NO emission was significantly higher than NO_2 emission, with NO accounting for approximately 80 % of NO_x (NO

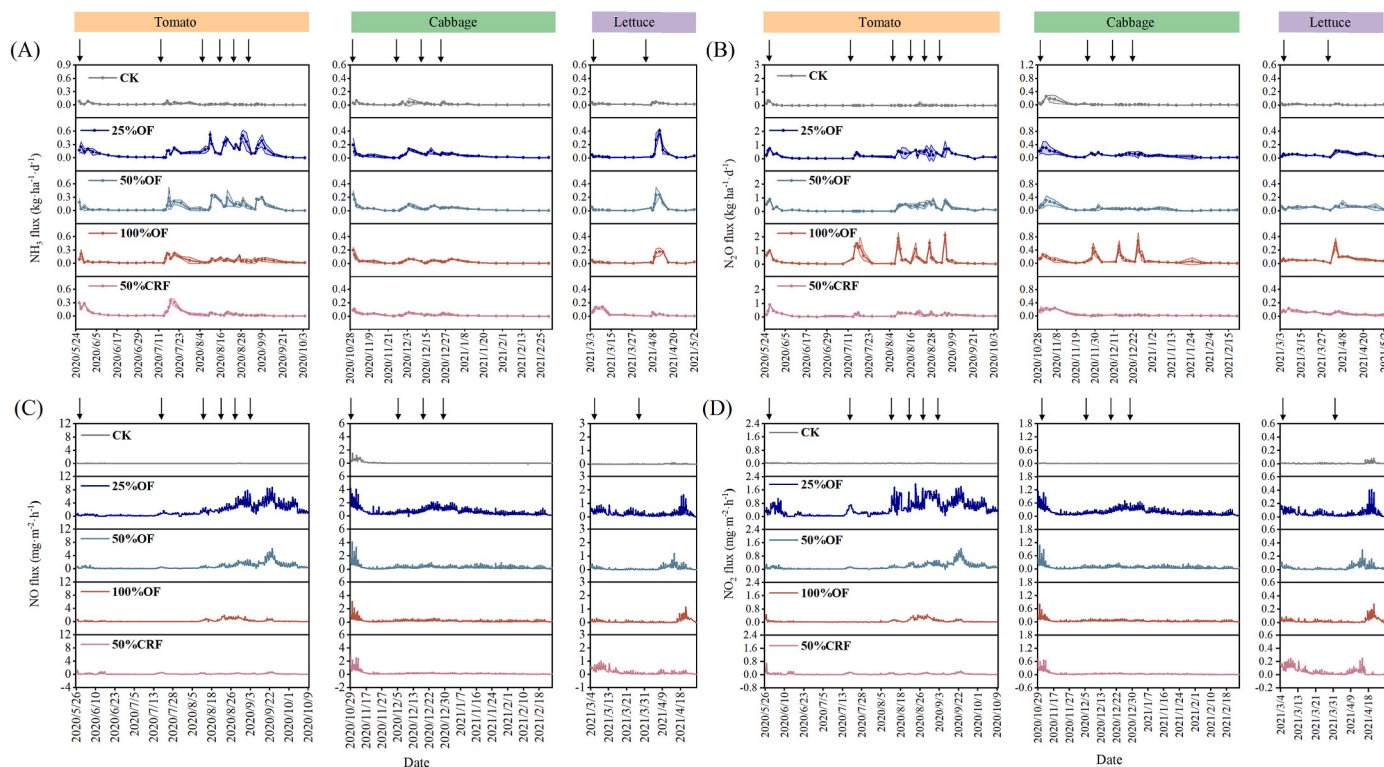


Fig. 1. NH_3 flux (A), N_2O flux (B), NO flux (C), NO_2 flux (D) under different treatments of greenhouse vegetable cultivation. The arrows indicate date of fertilizer application. The shadow indicates standard errors of three replicates.

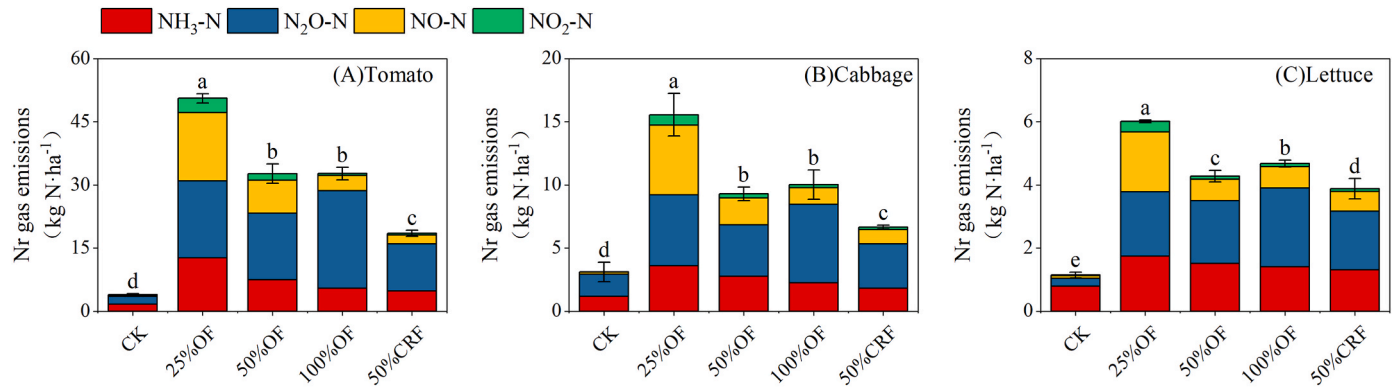


Fig. 2. NH_3 , N_2O , NO and NO_2 emissions under different treatments for tomato (A), cabbage (B) and tomato (C). Bars following columns represent standard errors of three replicates. Different lowercase letters indicate significant differences among five treatments at $P < 0.05$.

+ NO_2) emission (Table S2).

3.2. The total Nr gas emissions and mitigation effects

The total Nr gas emissions were 50.5, 32.5, 32.7 and 18.7 kg N-ha^{-1} , respectively under 25 %OF, 50 %OF, 100 %OF and 50 %CRF during tomato cultivation; 15.6, 9.3, 10.0 and 6.6 kg N-ha^{-1} during cabbage cultivation, respectively; and 5.9, 4.3, 4.7 and 3.7 kg N-ha^{-1} during lettuce cultivation, respectively (Fig. 3A). Compared with conventional fertilizer, the annual Nr gas emission was reduced by 18.6 % and 35.0 % for 50 %OF (organic substitution) and 50 %CRF (controlled-release fertilizer), respectively (Fig. 3A). For vegetable yield, controlled-release fertilizer showed high yield of tomato, cabbage and lettuce among all the treatments, but statistically significant differences were only detected between CK (no fertilization) and fertilized treatments (Table S3).

3.3. EFs and proportions of Nr gas emissions

The annual EFs of NH_3 , N_2O , NO and NO_2 (integrated across vegetable crops and N managements) were 0.6 % (0.3–1.1 %), 1.6 % (1.0–2.2 %), 0.8 % (0.3–1.8 %) and 0.2 % (0.0–0.4 %), respectively (Table 2). Controlled-release fertilizer exhibited the lowest EFs for the four gases (Table S4). Annual Nr gas losses accounted for 5.1 %, 3.0 %, 3.1 % and 1.7 % of N application rate, respectively under 25 %OF, 50 %OF, 100 %OF, 50 %CRF (Table 2). Tomatoes have higher EFs for Nr gases than cabbage and lettuce (Table S4). The proportion of N_2O emission (expressed as $\text{N}_2\text{O-N}$) accounted for the highest (36.0–67.3 %) among the four Nr gas emissions. The proportion of NO emission (32.7 %) in 25 %OF exceeded that of NH_3 volatilization' proportion (24.9 %),

Table 2

The EFs of NH_3 , N_2O , NO and NO_2 under different treatments of greenhouse vegetable cultivations.

Treatments	NH_3 EF (%)	N_2O EF (%)	NO EF (%)	NO_2 EF (%)	Nr gas EF (%)
25 %OF	1.1 %	1.8 %	1.8 %	0.4 %	5.1 %
50 %OF	0.7 %	1.4 %	0.8 %	0.1 %	3.0 %
100 %OF	0.4 %	2.2 %	0.4 %	0.1 %	3.1 %
50 %CRF	0.3 %	1.0 %	0.3 %	0.0 %	1.7 %
Average	0.6 %	1.6 %	0.8 %	0.2 %	3.2 %

while in other treatments, the proportion of NO emission was lower than that of NH_3 volatilization (Fig. 3B).

3.4. Microbial mechanisms underlying emission reductions

Compared to conventional fertilization, controlled-release fertilizer significantly reduced soil enzyme activities, with urease activity decreasing by 32.1 %, hydroxylamine reductase by 3.0 %, and denitrification enzyme by 45.2 %, while notably increasing protease activity by 19.4 % (Fig. 4). Increasing the proportion of organic fertilizer decreased soil urease activity, by 24.6 % with 100 %OF compared to conventional fertilizer (Fig. 4A). The application of controlled-release fertilizer significantly lowered the gene copy numbers of AOA *amoA*, *hao*, *nirK*, and *norB* by 32.3 %, 38.7 %, 62.4 %, and 66.0 %, respectively, compared to conventional fertilization. 100 %OF showed the highest gene expression levels of *narG*, *nirS*, *nirK*, and *norB*, with increases of 64.0 %, 16.3 %, 54.3 %, and 37.4 %, respectively (Fig. 5). Although increasing

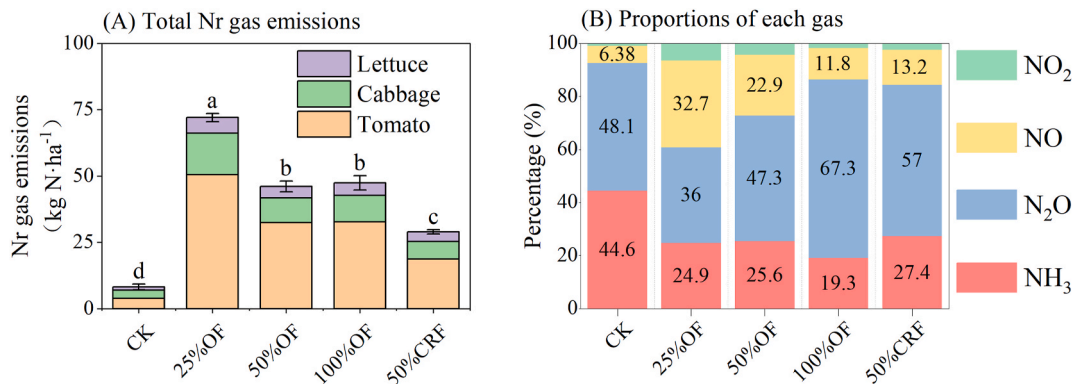


Fig. 3. The annual total Nr gas emissions in greenhouse systems (A), and composition of NH_3 , N_2O , NO , and NO_2 to annual Nr gas emissions (B). Bars following columns represent standard errors of three replicates. Different lowercase letters indicate significant differences among five treatments ($P < 0.05$). The labels were the absolute proportions of each gas to Nr gas emissions.

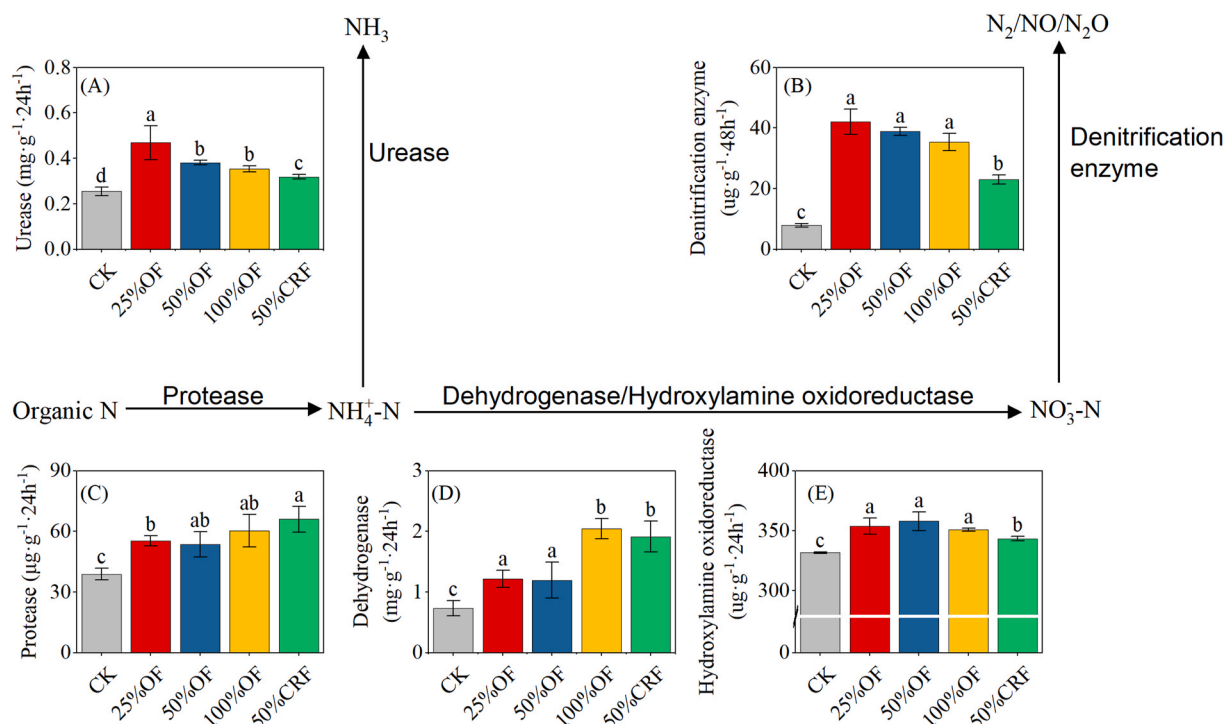


Fig. 4. The activities of urease (A), denitrification enzyme (B), protease (C), dehydrogenase (D) and hydroxylamine reductase (E) under different treatments. Bars following columns represent standard errors of three replicates. Different lowercase letters indicate significant differences among five treatments at $P < 0.05$.

the proportion of organic fertilizer didn't significantly affect the gene copy number of *nosZ*, the (*nirS* + *nirK*)/*norB* ratio was notably increased in 100 %OF treatment (Fig. 5K).

3.5. Driving factors of Nr gas emissions

The relative contributions of predictor variables to NH_3 , N_2O , NO, and NO_2 fluxes, as determined by the BRT model, were shown in Fig. 6. N application rate, soil temperature, soil moisture were the top three drivers of NH_3 volatilization in greenhouse vegetable production, with relative contributions of 19.5 %, 18.1 % and 13.9 %, respectively (Fig. 6A). Soil temperature, fertilizer type and N application rate were the top three drivers of N_2O emission, with relative contributions of 33.2 %, 29.0 % and 13.6 %, respectively (Fig. 6C). N application rate, soil temperature and fertilizer type were the top three drivers of NO and NO_2 emission, with relative contributions of 29.8 %, 20.5 % and 20.3 % for NO emission (Figs. 6E), and 33.4 %, 22.2 % and 20.8 % for NO_2 emission (Fig. 6G), respectively. The BRT model exhibited high predictive accuracy, explaining 84 %, 78 %, 94 % and 98 % of the variance for NH_3 volatilization, N_2O emission, NO emission and NO_2 emission (Fig. 6).

4. Discussion

4.1. Novelty of synchronous Nr gas monitoring and key emission patterns

This paper firstly reports the simultaneous emission characteristics of NH_3 , N_2O , NO, and NO_2 in greenhouse vegetable cultivation. Further, it also provides the EFs and proportions of these four gases and reveals underlying mechanisms of N management on Nr gas. Our findings provide a solid scientific foundation for reducing both Nr pollution and carbon emissions in greenhouse vegetable cultivation. Notably, in this study, the emission peak of NO continued for 3–15 days after fertilization, which was later than the peak of N_2O emission (Zhang et al., 2016). It was most likely due to anaerobic conditions of the soil surface after fertilization, where denitrification was stronger than nitrification

(Williams et al., 1992; Zhou et al., 2007). The *norB* gene copy number of nitric oxide reductase (NOR) exhibited a significant increase under high WFPS conditions, and NO was rapidly reduced to N_2O under high moisture and less oxygen available conditions (Barrat et al., 2022; Zhang et al., 2016). Soil aeration increased after water was leached underground, low moisture and good aeration were conducive to NO production (Cai et al., 2012), extending the duration of NO emission. Thus, to accurately assess total Nr gas losses, extensive and intensive monitoring must be conducted within 15 days after fertilization.

4.2. Mechanisms of Nr gas reduction under different N managements

Increasing the proportion of organic fertilizer can effectively reduce NH_3 volatilization and NO emission effectively (Han et al., 2021; Huang et al., 2016; Xia et al., 2017b). There is a positive correlation between the proportion of organic fertilizer replacing urea and the reduction of NH_3 volatilization (Liu et al., 2021), which can be attributed to the ability of organic fertilizer to increase the soil organic acid content, lower soil pH, improve soil CEC, enhance NH_4^+ absorption (Agbede and Oyewumi, 2022; Yang et al., 2020). It is generally believed that the application of urea increases soil urease activity, this study found that increasing the proportion of organic fertilizer effectively reduced soil urease activity, thereby inhibiting the conversion of ammonium nitrogen to NH_3 and reducing NH_3 volatilization. Additionally, the study revealed that the expression of the *nirK* gene, which directly influences NO production, was significantly lower in organic substitution compared to conventional fertilizer. This reduction in NO emissions associated with increasing proportions of organic fertilizer may be attributed to organic fertilizer providing sufficient organic N as a substrate, which promotes denitrification and reduces NO consumption. Furthermore, organic fertilizer slows SOM decomposition, resulting in increased N fixation during mineralization processes (Han et al., 2021; He et al., 2019; Li et al., 2023). These findings underscore the significant role of organic fertilizer in reducing NH_3 and NO emissions.

Organic fertilizer application increased soil organic carbon and nitrogen, which enhanced N_2O production through denitrification in the

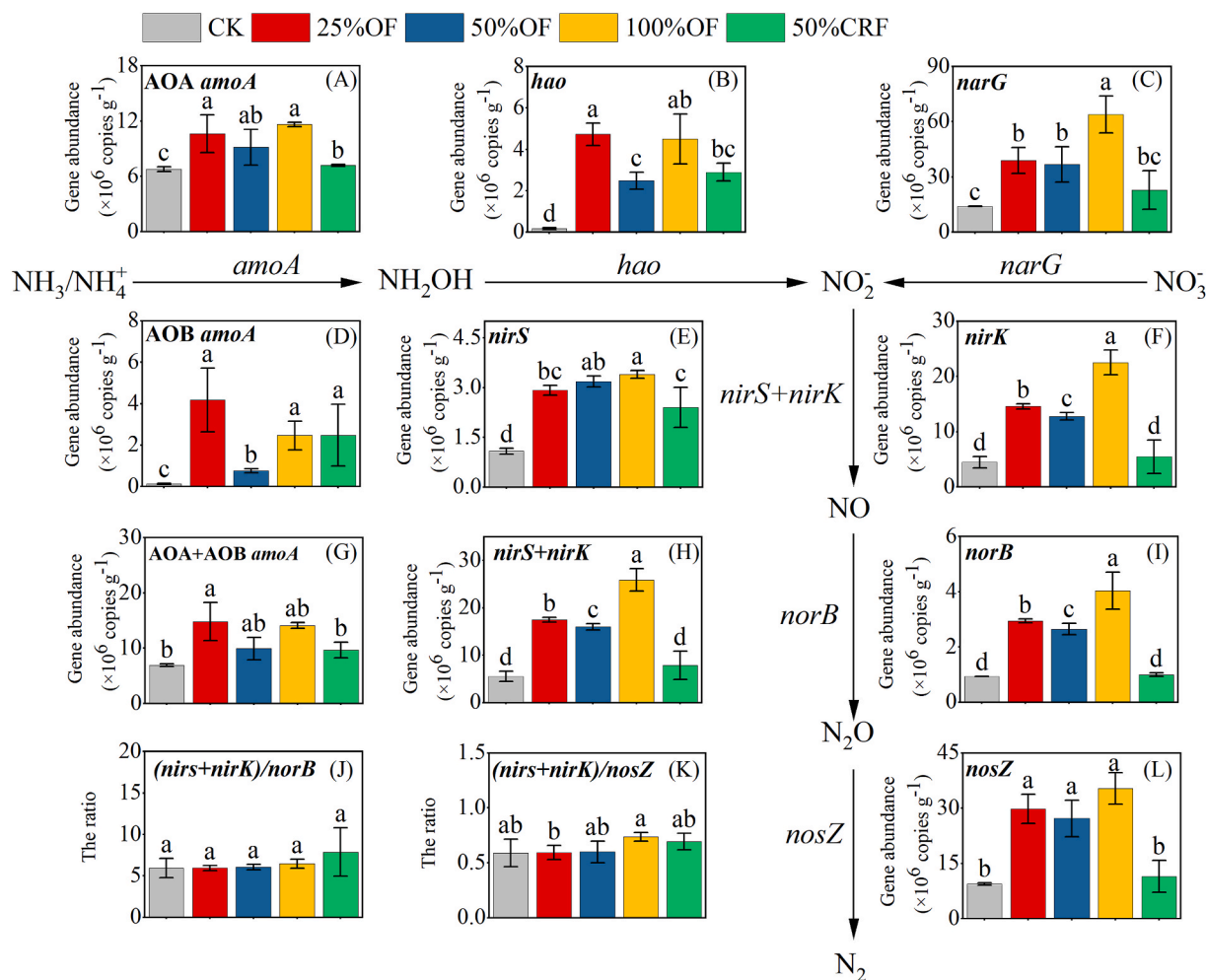


Fig. 5. The abundance of key genes involved in the N_2O emission, including AOA-*amoA* (A), AOB-*amoA* (D), *hao* (B), *narG* (C), *nirS* (E), *nirK* (F), *norB* (I) and *nosZ* (L) under different treatments. Bars following columns represent standard errors of three replicates. Different lowercase letters indicate significant differences among five treatments at $P < 0.05$.

soil (Hayakawa et al., 2009; Zhou et al., 2023). We observed the 100 % OF had particularly high N_2O emission peaks during topdressing with liquid vermicompost (Rizhiya et al., 2007), 100 %OF not only led to the highest expression levels of the *nirK* and *norB* genes but also exhibited the highest (*nirS* + *nirK*)/*nosZ* ratio. This ratio is commonly regarded as an indicator of the genetic potential for denitrification (Kong et al., 2021), indicating a significantly enhanced denitrification process of applying vermicompost. Although 100 %OF significantly affected the expression of the *nirK* and *norB* genes, it had no significant impact on the expression of the *amoA* and *hao* genes. This suggests that the primary mechanism by which the liquid vermicompost increases N_2O emissions is likely through promoting denitrification rather than nitrification. Vermicompost is rich in diverse amino acids and trace elements (Tian et al., 2021), and it can increase soil flora abundance, leading to improvements in both microaggregate structure and fertility (Jing et al., 2017). Many studies have emphasized the beneficial impact of vermicompost on vegetable yield and quality (Wang et al., 2017; Shang and Zhang, 2005), but no studies on the Nr gas emissions from applying vermicompost have been conducted. In addition, fertilizer type was found to be the second driving factor influencing N_2O emission flux (Fig. 6). For this reason, it is imperative to distinguish between the EFs of different fertilizer types to reduce uncertainty when estimating N_2O emissions from agricultural systems (Yao et al., 2017).

Controlled-release fertilizer inhibited the Nr gas emissions by reducing NH₃, NO and NO₂ emissions by decreasing urease and

denitrification enzyme activities and lowering expression levels of *amoA*, *hao*, *nirK* and *norB* genes during nitrification and denitrification processes (Wang et al., 2019). The reduction in NH₃ volatilization is attributed not only to decreased urease activity but also to its slow N release and low NH₄⁺ content in the soil (Zhao et al., 2013). Additionally, the reduced pool of NH₄⁺ can weaken the nitrification potential driving NO release (Liu et al., 2017; Sanchez-Martín et al., 2010). The effect of controlled-release fertilizer on NH₃ volatilization was more pronounced when the N application rate exceeded 200 kg N-ha⁻¹, compared to rates less than 200 kg N-ha⁻¹ (Zhang et al., 2019). Consequently, when compared to organic substitution, controlled-release fertilizer significantly reduced NH₃ volatilization for tomato (800 kg N-ha⁻¹) and cabbage (300 kg N-ha⁻¹), but not for lettuce (150 kg N-ha⁻¹). In terms of N_2O emission, controlled-release fertilizers can suppress the expression of genes associated with nitrification (*amoA* and *hao*) and denitrification (*nirK* and *norB*), indicating that both nitrification and denitrification play a role in reducing N_2O emissions of applying controlled-release fertilizer. While controlled-release fertilizer reduced N_2O emission by 38 % across all crops (Zhang et al., 2019), in this study, controlled-release fertilizer combined with organic fertilizer had no significant effects on reducing N_2O emission during cabbage and lettuce cultivation but significantly inhibited it during tomato cultivation. This difference can be attributed to the longer planting duration and higher frequency of topdressing applications for tomato cultivation than cabbage and lettuce. Moreover, it is worth noting that the majority of

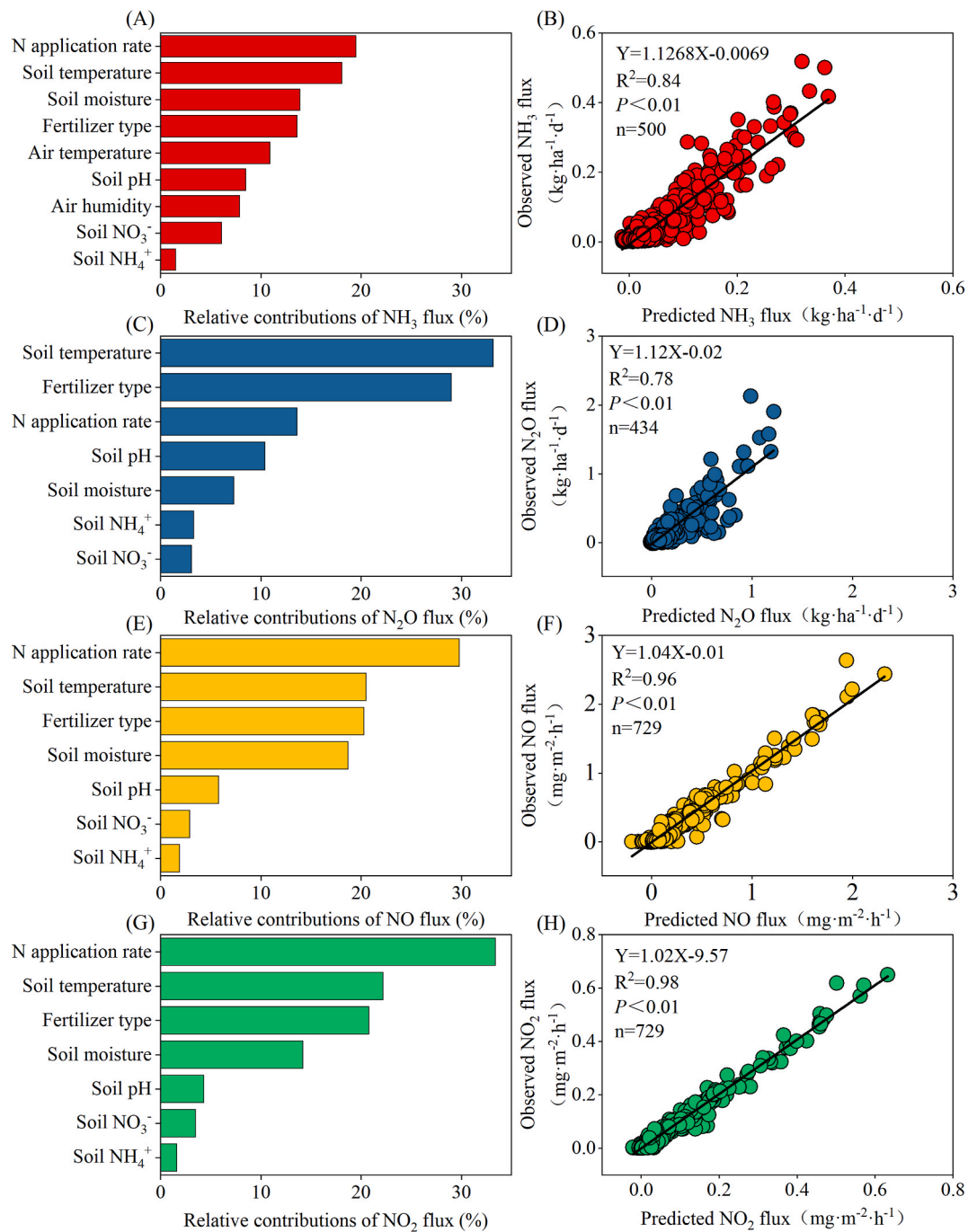


Fig. 6. Relative contribution (%) of predictor variables to NH₃ flux (A), N₂O flux (C), NO flux (E) and NO₂ flux (F) in the BRT model. The flux predicted by BRT model vs observed fluxes are shown for NH₃ flux (B), N₂O flux (D), NO flux (F) and NO₂ flux (G).

studies included in these meta-analyses focused on grain crops, with relatively limited data available regarding vegetable cultivation (Xia et al., 2017a; Zhang et al., 2019).

4.3. Comparison of EFs of NH₃, N₂O, NO and NO₂ in greenhouse cultivation

Among the four Nr gas emissions, the EFs and proportions of N₂O are the highest. The observed N₂O emissions from fertilized greenhouse vegetable cultivation ranged within 1.0–2.5 %, which was consistent with previous studies conducted in other districts within the same metropolitan area (e.g., Beijing) (Yan et al., 2014; Zhang et al., 2018), but higher than those reported in Southern China (1104 kg N·ha⁻¹) due

to the higher N application rate in northern China (1250 kg N·ha⁻¹ in this study) (Min et al., 2012). All mentioned studies consistently indicated significantly higher levels of N₂O EFs from greenhouse vegetable fields compared to wheat-maize fields (0.11 %–0.22 %) in the same region (Liao et al., 2020; Wang et al., 2023). Thus, it can be concluded that long-term inappropriate fertilizer application changes greenhouse vegetable fields into an expanding N pool that increases N₂O emission.

The EF of NH₃ for greenhouse cultivation is lower than that of N₂O indicating a shift in the dominant Nr loss pathway compared to the open-field system. Conversely, the EF of NH₃ (10.7–13.1 %) for open-field is markedly higher than that of N₂O (0.7 %–2.4 %) (Zheng et al., 2018). The NH₃ EFs inside greenhouses are significantly lower than open-field vegetable systems (Miao et al., 2022). This discrepancy is

largely attributable to the semi-enclosed environment of greenhouses, the sluggish air circulation and long NH_3 retention time in greenhouses allow the plant canopy to absorb more portions of NH_3 volatilized from the soil (Walker et al., 2013). NH_3 is highly soluble in water, and elevated humidity inside the greenhouse promotes dissolution of volatilized NH_3 into water films on surfaces, followed by dry and wet deposition back to soil. This process effectively recycles NH_3 inside greenhouse, enriching the soil inorganic N content and increasing N_2O emission (Ding et al., 2022; Miao et al., 2022). Additionally, consistent drip irrigation maintains high soil moisture (WFPS >60 %) (Fig. S3), creating anaerobic microsites that favor denitrification and further amplify N_2O emissions (Zhang et al., 2016).

According to earlier research, NO and N_2O emissions may exhibit a competitive relationship, where higher soil moisture content corresponded to consume NO emissions but increase N_2O emissions (Zhang et al., 2016). This aligns with findings that high soil water-filled pore space (WFPS) promotes denitrification-dominated N_2O production while suppressing nitrification-driven NO release (Akiyama and Tsuruta, 2003). However, they can also demonstrate synergistic behavior or increase simultaneously under specific conditions (Zhou et al., 2007), such as after organic fertilizer application where low C/N ratios enhance both gases via rapid mineralization (Akiyama and Tsuruta, 2003). NO emissions exhibit a longer duration of elevated concentration compared to N_2O (Fig. S5), accounting for 11.0 %–35.5 % of the Nr gas emissions. NO_2 accounts for only a small fraction of Nr gas in greenhouse vegetable systems. In fact, several studies have reported negative NO_2 fluxes, indicating NO_2 uptake by the soil, a phenomenon that has been consistently observed across different environments (Akiyama and Tsuruta, 2002, 2003; Yamulki et al., 1997). Notably, EF of NO emission from vegetable cultivation (1.71 %) was highlighted among croplands (Liu et al., 2017). Consequently, the NO emission in greenhouse vegetable fields requires particular attention, and further investigation is warranted to elucidate the interaction mechanisms of NO and N_2O emissions (Stehfest and Bouwman, 2006).

5. Conclusion

Our study is the first to synchronously monitor the NH_3 , N_2O , NO and NO_2 emissions in greenhouse vegetable cultivation. The annual Nr gas loss was observed to be 72.0, 46.1, 47.4, and 29.0 kg N·ha⁻¹ under the treatments of 25 %OF, 50 %OF, 100 %OF and 50 %CRF, respectively. The duration of NO and NO_2 emission peak (3–15d) was longer than those of N_2O and NH_3 (3–8d). The EFs of NH_3 , N_2O , NO and NO_2 varied from 0.2–1.4 %, 0.6–2.7 %, 0.2–2.0 % and 0.0–0.4 %, respectively. Among the four gas emissions, N_2O emission had the highest proportion (36.0–67.3 %), while NH_3 and NO accounted for 19.3–44.6 % and 6.4–32.7 %, respectively. Controlled-released N combined with organic fertilizer has the most effective reduction of Nr gas emissions by lowering urease activity and expressions of *amoA*, *hao*, *nirK* and *norB* genes, followed by increasing the proportion of organic fertilizer. The main factors influencing NH_3 volatilization were N application rate, soil moisture and soil temperature, while the main factors influencing N_2O , NO and NO_2 emissions were N application rate, fertilizer type and soil temperature. This study offered solid evidence of the EFs and proportions of NH_3 , N_2O , NO and NO_2 for a more accurate Nr accounting, and gave N management solutions for the reduction of pollution and carbon emissions in greenhouse vegetable cultivation.

CRediT authorship contribution statement

Tiantian Miao: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Data curation. **Bin Wang:** Writing – review & editing, Methodology, Investigation, Funding acquisition. **Yunfan Wan:** Methodology, Investigation. **Yu'e Li:** Investigation, Funding acquisition. **Tianjing Ren:** Methodology, Investigation, Data curation. **Huan Liu:** Writing – review & editing, Software.

Jiaqi Zhang: Software, Investigation. **Andong Cai:** Methodology, Investigation, Data curation. **Waseem Hassan:** Writing – review & editing.

Declaration of competing interest

We declare that the authors have no conflicts of interest to this work. We also declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2025.127081>.

Data availability

Data will be made available on request.

References

- Agbede, T.M., Oyewumi, A., 2022. Benefits of biochar, poultry manure and biochar–poultry manure for improvement of soil properties and sweet potato productivity in degraded tropical agricultural soils. *Resources, Environment and Sustainability* 7, 100051.
- Akiyama, H., Tsuruta, H., 2002. Effect of chemical fertilizer form on N_2O , NO and NO_2 fluxes from Andosol field. *Nutrient Cycl. Agroecosyst.* 63, 219–230.
- Akiyama, H., Tsuruta, H., 2003. N_2O , NO and NO_2 fluxes from soils after the application of manures and urea. *J. Environ. Qual.* 32, 423–431.
- Barrat, H.A., Charteris, A.F., Le Cocq, K., Abadie, M., Clark, I.M., Chadwick, D.R., Cardenas, L., 2022. N_2O hot moments were not driven by changes in nitrogen and carbon substrates or changes in N cycling functional genes. *Eur. J. Soil Sci.* 73 (1).
- Bi, R., Xu, X., Wang, B., Jiao, Y., Zhang, Q., Xiong, Z., 2024. Evidence of complete ammonia-oxidizing microbial communities and their contribution to N_2O emissions in typical vegetable fields across China. *Soil Biol. Biochem.* 194, 109423.
- Cai, A., Han, T., Ren, T., Sanderman, J., Rui, Y., Wang, B., Smith, P., Xu, M., Li, Y., 2022. Declines in soil carbon storage under no tillage can be alleviated in the long run. *Geoderma* 425, 116028.
- Cai, Y.J., Ding, W.X., Xiang, J., 2012. Factors controlling N_2O and NO emissions from agricultural soils and their influencing mechanisms: A review. *Soils* 44, 881–887.
- Deng, N., Gubry-Rangin, C., Song, X.T., Ju, X.T., Liu, S.Y., Shen, J.P., Di, H.J., Han, L.L., Zhang, L.M., 2024. AOB *Nitrosospora* cluster 3a.2 (D11) dominates N_2O emissions in fertilised agricultural soils. *J. Environ. Manag.* 355, 120504.
- Ding, W.H., Zhang, G.L., Yi, X.P., Kang, J.H., Zhang, J., Li, G.C., Zhang, J.F., Chang, N.J., Li, H., 2023. Improving potential of reactive nitrogen and carbon footprint of intensified greenhouse cucumber–tomato production with optimized drip irrigation with nitrogen reduction strategies. *J. Clean. Prod.* 424, 138898.
- Ding, W.H., Chang, N.J., Zhang, J., Li, G., Zhang, J.F., Ju, X.H., Zhang, G.L., Li, H., 2022. Optimized fertigation mitigates N_2O and NO emissions and enhances NH_3 volatilizations in an intensified greenhouse vegetable system br. *Agric. Water Manag.* 272.
- Elith, J., Leathwick, J.R., Hastie, T., 2008. A working guide to boosted regression trees. *J. Anim. Ecol.* 77, 802–813.
- Han, Z., Wang, J., Xu, P., Li, Z., Liu, S., Zou, J., 2021. Differential responses of soil nitrogen-oxide emissions to organic substitution for synthetic fertilizer and biochar amendment in a subtropical tea plantation. *GCB Bioenergy* 13, 1260–1274.
- Hao, X.Y., Gao, W., Wang, Y.J., Jin, J.Y., Huang, S.W., Tang, J.W., Zhang, Z.Q., 2012. Effects of combined application of organic manure and chemical fertilizers on ammonia volatilization from greenhouse vegetable soil. *Sci. Agric. Sin.* 45, 4403–4414.
- Hayakawa, A., Akiyama, H., Sudo, S., Yagi, K., 2009. N_2O and NO emissions from an Andisol field as influenced by pelleted poultry manure. *Soil Biol. Biochem.* 41, 521–529.
- He, T., Yuan, J., Luo, J., Wang, W., Fan, J., Liu, D., Ding, W., 2019. Organic fertilizers have divergent effects on soil N_2O emissions. *Biol. Fertil. Soils* 55, 685–699.

- Huang, S., Lv, W., Bloesies, S., Shi, Q., Pan, X., Zeng, Y., 2016. Effects of fertilizer management practices on yield-scaled ammonia emissions from croplands in China: a meta-analysis. *Field Crops Res.* 192, 118–125.
- Jing, D.W., Wang, M.Y., Zhang, H., Li, S.P., 2017. Effects of Vermicompost Co-applied with Urea on Root Characteristics and Humus in Rhizosphere Soil of Cowpea, 48. *Transactions of the Chinese Society for Agricultural Machinery*, pp. 212–219.
- Ju, X.T., Kou, C.L., Christie, P., Dou, Z.X., Zhang, F.S., 2007. Changes in the soil environment from excessive application of fertilizers and manures to two contrasting intensive cropping systems on the North China Plain. *Environ. Pollut.* 145, 497–506.
- Kong, D., Jin, Y., Yu, K., Swaney, D.P., Liu, S., Zou, J., 2021. Low N₂O emissions from wheat in a wheat-rice double cropping system due to manure substitution are associated with changes in the abundance of functional microbes. *Agric. Ecosyst. Environ.* 311, 107318. <https://doi.org/10.1016/j.agee.2021.107318>.
- Li, M., Peng, J., Lu, Z., Zhu, P., 2023. Research progress on carbon sources and sinks of farmland ecosystems. *Resources, Environment and Sustainability* 11, 100099.
- Li, T.L., Qi, M.F., Meng, S.D., 2022. Sixty years of facility horticulture development in China: achievements and prospects. *Acta Hortic. Sin.* 49, 2119–2130.
- Liao, X., Niu, Y., Liu, D., Chen, Z., He, T., Luo, J., Lindsey, S., Ding, W., 2020. Four-year continuous residual effects of biochar application to a sandy loam soil on crop yield and N₂O and NO emissions under maize-wheat rotation. *Agric. Ecosyst. Environ.* 302, 107109.
- Liu, B., Wang, X., Ma, L., Chadwick, D., Chen, X., 2021. Combined applications of organic and synthetic nitrogen fertilizers for improving crop yield and reducing reactive nitrogen losses from China's vegetable systems: a meta-analysis. *Environ. Pollut.* 269, 116143.
- Liu, S.W., Lin, F., Wu, S., Ji, C., Sun, Y., Jin, Y.G., Li, S.Q., Li, Z.F., Zou, J.W., 2017. A meta-analysis of fertilizer-induced soil NO and combined NO+N₂O emissions. *Glob. Change Biol.* 23, 2520–2532.
- Miao, T.T., Wang, B., Cai, A.D., Ren, T.J., Wan, Y.F., Meng, Y., Li, Y.E., 2022. Large differences in ammonia emission factors between greenhouse and open-field systems under different practices across Chinese vegetable cultivation. *Sci. Total Environ.* 852.
- Min, J., Shi, W., Xing, G., Powlson, D., Zhu, Z., 2012. Nitrous oxide emissions from vegetables grown in a polytunnel treated with high rates of applied nitrogen fertilizers in Southern China. *Soil Use Manag.* 28, 70–77.
- Qin, K., Rao, L.L., Xu, J., Bai, Y., Zou, J.H., Hao, N., Li, S.S., Yu, C., 2017. Estimating ground level NO₂ concentrations over central-Eastern China using a Satellite-based Geographically and temporally weighted regression model. *Remote Sens.* 9.
- Quick, A.M., Reeder, W.J., Farrell, T.B., Tonina, D., Feris, K.P., Benner, S.G., 2019. Nitrous oxide from streams and rivers: a review of primary biogeochemical pathways and environmental variables. *Earth Sci. Rev.* 191, 224–262.
- Rizhiya, E., Bertora, C., van Vliet, P.C.J., Kuikman, P.J., Faber, J.H., van Groenigen, J.W., 2007. Earthworm activity as a determinant for N₂O emission from crop residue. *Soil Biol. Biochem.* 39 (8), 2058–2069.
- Sanchez-Martín, L., Meijide, A., García-Torres, L., Vallejo, A., 2010. Combination of drip irrigation and organic fertilizer for mitigating emissions of nitrogen oxides in semiarid climate. *Agric. Ecosyst. Environ.* 137, 99–107.
- Shan, L., He, Y., Chen, J., Huang, Q., Wang, H., 2015. Ammonia volatilization from a Chinese cabbage field under different nitrogen treatments in the Taihu Lake Basin, China. *J. Environ. Sci.* 38, 14–23.
- Shang, Q.M., Zhang, Z.G., 2005. Physical and chemical properties of wormcast and its effect on plug seedling of tomato. *China Veg.* 10–12.
- Shuai, Y.J., Liu, T.Q., Cao, C.G., Li, C.F., 2018. Effect of combined application of organic and inorganic fertilizers on soil nitrous oxide emission in maize-cowpea systems in central China. *Chin. J. Eco-Agric.* 26, 1763–1772.
- Stehfest, E., Bouwman, L., 2006. N₂O and NO emission from agricultural fields and soils under natural vegetation: summarizing available measurement data and modeling of global annual emissions. *Nutrient Cycl. Agroecosyst.* 74, 207–228.
- Tian, T., Tian, Y.Q., Gao, L.H., 2021. Research progress on soil quality of protected vegetable fields. *China Veg.* 35–44.
- Walker, J.T., Jones, M.R., Bash, J.O., Myles, L., Meyers, T., Schwede, D., Herrick, J., Nemitz, E., Robarge, W., 2013. Processes of ammonia air-surface exchange in a fertilized Zea mays canopy. *Biogeosciences* 10, 981–998.
- Wang, B., Li, J., Wan, Y., Li, Y.e., Qin, X., Gao, Q., Waqas, M.A., Wilkes, A., Cai, W., You, S., Zhou, S., 2018. Responses of yield, CH₄ and N₂O emissions to elevated atmospheric temperature and CO₂ concentration in a double rice cropping system. *Eur. J. Agron.* 96, 60–69.
- Wang, W., Li, Y., Wang, W., Ding, K., Zhang, K., Zhang, Y., Liu, K., Li, J., Li, D., Lian, R., Li, D., Gu, J., Li, J., 2024. Global meta-analysis of individual and combined nitrogen inhibitors: enhancing plant productivity and reducing environmental losses. *Glob. Change Biol.*
- Wang, X., Wang, S., Zang, H., Nie, J., Zhao, J., Wang, P., Peixoto, L., Yang, Y., Olesen, J. E., Zeng, Z., 2023. Replacing chemical fertilizer with manure reduces N₂O emissions in winter wheat – summer maize cropping system under limited irrigation. *J. Environ. Manag.* 336, 117677.
- Wang, X.X., Zhao, F.Y., Zhang, G.X., Zhang, Y.Y., Yang, L.J., 2017. Vermicompost improves tomato yield and quality and the biochemical properties of soils with different tomato planting history in a greenhouse study. *Front. Plant Sci.* 8.
- Wang, X.X., Cao, B., Liang, H.S., Wang, J.C., Chen, Y.H., Liu, D.S., Yue, J.Q., 2019. Combined application of controlled-release nitrogen fertilizer and water-soluble fertilizer to reduce N₂O emission in greenhouse soil. *Journal of Plant Nutrition and Fertilizers* 25 (12), 2084–2094.
- Wang, Y., Yao, Z., Zheng, X., Subramaniam, L., Butterbach-Bahl, K., 2022. A synthesis of nitric oxide emissions across global fertilized croplands from crop-specific emission factors. *Glob. Change Biol.* 28, 4395–4408.
- Wang, Y.C., Wang, L.Y., Gao, Z.L., Ma, W.Q., 2016. Effects of applying organic manure and chemical fertilizers on N₂O emissions from tomato greenhouse soil. *Journal of Hebei Agricultural University* 39, 1–6.
- Wang, Z.H., Liu, X.J., Ju, X.T., Zhang, F.S., 2002. Field in situ determination of ammonia volatilization from soil: venting method. *Journal of Plant Nutrition and Fertilizers* 205–209.
- Williams, E.J., Hutchinson, G.L., Fehsenfeld, F.C., 1992. NO_x and N₂O emissions from soil. *Glob. Biogeochem. Cycles* 6, 351–388.
- Xia, L., Lam, S.K., Chen, D., Wang, J., Tang, Q., Yan, X., 2017a. Can knowledge-based N management produce more staple grain with lower greenhouse gas emission and reactive nitrogen pollution? A meta-analysis. *Glob. Change Biol.* 23, 1917–1925.
- Xia, L., Lam, S.K., Yan, X., Chen, D., 2017b. How does recycling of livestock manure in agroecosystems affect crop productivity, reactive nitrogen losses, and soil carbon balance? *Environ. Sci. Technol.* 51, 7450–7457.
- Xu, X., Bi, R., Song, M., Wang, B., Dong, Y., Zhang, Q., Xiong, Z., 2024. Optimizing organic substitution: balancing carbon sequestration and priming effects of a six-year field experiment for sustainable vegetable production. *Sustain. Prod. Consum.* 44, 14–24.
- Yamulki, S., Harrison, R.M., Goulding, K.W.T., Webster, C.P., 1997. N₂O, NO and NO₂ fluxes from a grassland: effect of soil pH. *Soil Biol. Biochem.* 29, 1199–1208.
- Yan, H., Xie, L., Guo, L., Fan, J., Diao, T., Lin, M., Zhang, H., Lin, E., 2014. Characteristics of nitrous oxide emissions and the affecting factors from vegetable fields on the North China Plain. *J. Environ. Manag.* 144, 316–321.
- Yan, S., Yu, Y.Y., Liang, J.F., Guo, N., Liu, Y., Zhao, K.L., Liu, L., 2023. Preliminary study on fertilization and soil fertility of facility vegetable fields in Beijing. *Soil Fertil. Sci. China* 125–133.
- Yang, J.G., Zhang, P.F., Ni, X.H., Xiao, Q., Cao, B., Zou, G.Y., Liu, B.C., 2014. Effects of controlled release fertilizer on soil nitrate leaching, N₂O emission and fruit yield and quality in greenhouse tomato production system. *Journal of Agro-environment Science* 33, 1849–1857.
- Yang, Q., Liu, P., Dong, S., Zhang, J., Zhao, B., 2020. Combined application of organic and inorganic fertilizers mitigates ammonia and nitrous oxide emissions in a maize field. *Nutrient Cycl. Agroecosyst.* 117, 13–27.
- Yao, Z., Yan, G., Zheng, X., Wang, R., Liu, C., Butterbach-Bahl, K., 2017. Reducing N₂O and NO emissions while sustaining crop productivity in a Chinese vegetable-cereal double cropping system. *Environ. Pollut.* 231, 929–941.
- Yi, Z., Zhang, Z., Chen, G., Rengel, Z., Sun, H., 2023. Microplastics have rice cultivar-dependent impacts on grain yield and quality, and nitrogenous gas losses from paddy, but not on soil properties. *J. Hazard Mater.* 446, 130672.
- Zhang, J., Li, H., Wang, Y., Deng, J., Wang, L., 2018. Multiple-year nitrous oxide emissions from a greenhouse vegetable field in China: effects of nitrogen management. *Sci. Total Environ.* 616–617, 1139–1148.
- Zhang, W., Liang, Z., He, X., Wang, X., Shi, X., Zou, C., Chen, X., 2019. The effects of controlled release urea on maize productivity and reactive nitrogen losses: a meta-analysis. *Environ. Pollut.* 246, 559–565.
- Zhang, Y.Y., Mu, Y.J., Zhou, Y.Z., Tian, D., Liu, J.F., Zhang, C.L., 2016. NO and N₂O emissions from agricultural fields in the North China Plain: origination and mitigation. *Sci. Total Environ.* 551, 197–204.
- Zhao, B., Dong, S., Zhang, J., Liu, P., 2013. Effects of controlled-release fertiliser on nitrogen use efficiency in summer maize. *PLoS One* 8, e70569.
- Zheng, L., Wang, X.D., Guo, L.P., Zhang, X.Y., Wang, D.Y., Niu, X.G., Yun, A.P., Li, Y.C., 2018. Impact of fertilization on ammonia volatilization and N₂O emissions in an open vegetable field. *Chin. J. Appl. Ecol.* 29, 4063–4070.
- Zhou, P., Bai, X., Wang, H., Yang, M., Bao, L., Deng, X., Chen, Z., Zhou, J., 2023. Optimizing nitrogen and water management for sustainable greenhouse vegetable production with less greenhouse gas emissions. *Agric. Ecosyst. Environ.* 352, 108529.
- Zhou, Z.X., Zheng, X.H., Wang, M.X., Butterbach-Bahl, K., 2007. CH₄, N₂O and NO emissions from a rice-wheat rotation cropping field in east China. *Climatic and Environmental Research* 12 (6), 751–760.